

Federal Bank: Building outstanding customer journeys with an advanced AI virtual assistant

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Federal Bank built an advanced AI personal assistant, gaining 100% response and boosting customer satisfaction by 20% to trust for complex banking transactions with Dialogflow.

About Federal Bank

Federal Bank Limited is a major digital-first Indian commercial bank with more than 1,250 branches and a customer base of more than 10 million across different states. As India's first bank to digitize all branches, Federal Bank has been a pioneer in leveraging tech to drive its business ever since it bought its first server in the 1990s. Now it is embracing cloud and AI technologies to forge the next chapter of India's digital banking journey. Federal Bank has been recognized as a Great Place to Work in a study conducted by the Great Place to Work Institute.

Google Cloud results

- Builds trust in virtual personal assistant with 98% accuracy in answers to customer queries
- Enables virtual banking to scale by more than doubling handled queries to 1.4 million per year
- Enhances business growth by enabling 50% savings in customer care costs through AI automation by 2025
- Frees up developers

25% rise in customer satisfaction
Dialog

Industries: Financial Services & Insurance

Location: India



About Riafy Technologies

Riafy Technologies builds AI consumer solutions for partners around the world, enabling them to ideate, build, and grow products collaboratively. Specializing in conversational AI, Riafy's products and solutions are used by 36 million people in 27 languages, with a global footprint spanning 160 countries. Riafy's work on Android and AI has been showcased at multiple Google I/Os.

products collaboratively. Specializing in Google Cloud (<https://cloud.google.com/>) and

Dialogflow Enterprise Edition (<https://cloud.google.com/dialogflow-enterprise/>) spanning 160 countries. Riafy's work on Android and AI has been showcased at multiple Google I/Os.

for creative
tasks by
saving five
hours per
day on
mundane
tasks

- Paves way
to complex
virtual
banking
transactions
through
trust built
by accurate,
human-
friendly
solution

For Federal Bank's (<https://www.federalbank.co.in/>) 10-million-plus customers, the face of India's first bank to digitize all branches is a friendly "boy" called Feddy (<https://www.federalbank.co.in/feddy>). The AI personal assistant introduced in 2020 has become such a hit, thanks to accurate and natural responses to queries, that he's now the trusted mascot for all of the bank's advertisements and social media.

Feddy is the culmination of an ambitious AI overhaul spearheaded by Jithesh PV, Federal Bank's Vice President and Head of Digital, to transform the bank's customer experience with a virtual agent that would "change the face of the bank in the next five years."

"Our dream was to build a powerful conversational AI platform from scratch with a different dimension than just buying an off-the-shelf solution and fitting applications around it. When Riafy introduced us to Dialogflow, we discovered infinite possibilities to reimagine the customer experience through a truly interactive, auto-learning virtual agent."

*—Jithesh PV, Head of
Digital, Federal Bank*

Jithesh sought more than a conventional chatbot that "ticked the box" of having an AI solution. Instead he envisioned a fully interactive customer partner that understood conversational context, and eventually would handle complex banking transactions, such as blocking payments or selecting a new debit card from a variety of options.

Federal Bank experimented with several conversational AI solutions and found that Dialogflow met all of the requirements in terms of accuracy, versatility, and contextual understanding. That's when the bank tapped Riafy Technologies (<https://www.riafy.me/tech/>) as a partner, and unlocked the power of Dialogflow (/dialogflow) to gain the lifelike conversational AI needed that turned Feddy into a virtual agent beloved by millions across India.

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**Enabling lifelike customer care by
capturing conversational nuance
and context**

Before being introduced to Dialogflow by Riafy, Jithesh's team tested roughly 30 conversational AI providers and encountered the same problems time and again. One major challenge involved handling basic queries expressed in a narrow wording range. For example, the solution might be able to answer "What is my bank balance?" but be stumped when asked "How much money do I have?"

The chatbots also tended to be one-size-fits-all, meaning they were impossible to develop strategically according to Federal Bank's vision of intelligent customer interaction. And the bots all required an enormous amount of time to be trained, distracting Jithesh's engineering team from the creative task of solution building.

"There was no automatic learning in the solutions we tested, and we had to train the bot for every question," recalls Jithesh. "We had to train roughly a thousand questions per month, and fine-tune questions for customers based on their level of understanding. That changed overnight when we adopted Dialogflow, and the platform trained itself."

Riafy came armed with seven years of research and development experience in AI consumer solutions. That meant Dialogflow's contextual speech understanding could combine with Riafy's own proprietary AI tool TextMagic, an intelligent self-learning tool that enables highly contextual and relevant information, to enable Feddy to engage in surprisingly natural and accurate conversation.

Dialogflow and TextMagic offered advanced AI conversational techniques such as mirroring, in which the bot takes on the customer's speech patterns, to build a sense of trust and demonstrate that the AI truly grasps what the customer needs.

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Since launching Feddy in 2020, Federal Bank has achieved significant measurable results in improving the customer journey. The platform's contextual algorithms driven by Dialogflow enable Feddy to choose the best answer to any question from a list of a thousand possibilities, with a 98% accuracy rate, says Jithesh.

Thanks to Dialogflow's contextual auto-learning capabilities, Feddy's ML engine does not need to be constantly trained to answer questions, saving Jithesh's development team up to five hours per day in mundane tasks, time that can then be devoted to inventing new AI solutions.

Today Feddy fields up to 1.4 million queries per year, compared to 600,000 with an earlier prototype solution from a different provider. The bank is on track to double the number of queries by 2025. With repeat customers to Feddy growing, Jithesh estimates the bot has driven a 25% rise in customer satisfaction in overall customer care.

The numbers translate into the most important outcome of Feddy's success: trust. As customers begin to rely on Feddy more than human agents, trust is being built that will enable the final stage of the bank's conversational AI journey: tasking Feddy with complex customer transactions, such as sending remittances, halting payments, or ordering new cards, without the help of a human agent.

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Architecting a multistage conversational AI strategy with a trusted partner

Federal Bank and Riafy worked in close collaboration to engineer Feddy's rollout in three stages based on the complexity of tasks required. Stage one was handling static queries such as information on how to open an account, in a natural conversational style that captures customer intent in a contextual manner.

Stage two was integration with Google Business Messages, an asynchronous mobile conversational experience that links a brand website with Google Maps and Search to engage customers and drive business. In asynchronous messaging, customers message a business at their own convenience, and come back to the conversation whenever they wish.

The third and most ambitious phase is enabling Feddy to handle transactional banking and is

expected to debut in March. Work is being completed on the AI "brain" supported by Dialogflow that will allow Feddy to take the customer through complex transactions without human help.

This last phase, says Jithesh, promises to transform Federal Bank's ability to automate banking operations and drive future growth and profitability. By 2025, he expects Feddy to be executing up to 4 million transactions, with 50% of contact center volume migrated to the chatbot, yielding 50% savings in customer relations costs.

Driving personalized banking solutions through Google machine learning tools

The story of Feddy's evolution is only beginning with the third phase of the chatbot's rollout. Next, Federal Bank plans to enable Feddy to be the front man for individualized customer relations management, offering personalized banking suggestions based on customer data.

To enable this journey, Jithesh plans to incorporate more Google Cloud AI and ML solutions such as BigQuery ML into the platform, to build the powerful data analytics engines required for individualized customer experiences that anticipate client needs.

"Once Feddy's customer engagement ecosystem is completed, opportunities arise for all types of cross-selling, up-selling, and lead management," says Jithesh. "To enable that, I expect use of Google services to double or triple in the next couple of

years, as our analytics team and marketing team invest in new analytics tools.”